

LESSON 06

ESP32 Fundamentals

<http://www.youtube.com/watch?v=mydCDdnSo5o&utm>

A connected microcontroller for robotics and IoT.

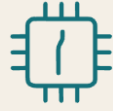


THE BIG IDEA

ESP32 brings more computing and connectivity to small projects

A beginner-friendly board can sense, decide, and communicate from one small package.

Think: local control + wireless connection.



More processing

High processing capability for its size supports robotics, automation, and IoT.



Built-in Wi-Fi

2.4 GHz Wi-Fi connects projects to networks and cloud services.



Built-in Bluetooth

Bluetooth links nearby phones, sensors, and controllers.

CHOOSE BY THE BEGINNER PROBLEM

ESP32 and Arduino UNO solve different beginner problems

Typical reference values; exact GPIO and ADC availability depends on the specific board.

FEATURE	ESP32	ARDUINO UNO R3
Processor	32-bit dual-core Xtensa LX6	8-bit ATmega328P
Clock speed	Up to 240 MHz	16 MHz
Wi-Fi	Built in, 2.4 GHz	None
Bluetooth	Built in on classic ESP32	None
GPIO	Many digital GPIO; board-dependent	14 digital I/O, 6 PWM
ADC	Up to 12-bit, board-dependent	6 channels, 10-bit
Operating voltage	3.3 V logic	5 V logic
Typical applications	Robotics, IoT, connected sensors	Simple control, learning, basic prototypes

Takeaway: choose ESP32 when connectivity, speed, and peripherals matter; choose UNO when simple 5 V projects and first-step learning matter.

A STEP UP FROM WI-FI ONLY

ESP32 has more capability than ESP8266.

Both boards provide built-in 2.4 GHz Wi-Fi. ESP32 adds more room for control, peripherals, and connected behavior.

CAPABILITY	ESP32	ESP8266
Processing	More processing headroom; dual-core on classic ESP32	Single-core, simpler workload
Wi-Fi	Built in	Built in
Bluetooth	Built in on classic ESP32	Not built in
Hardware peripherals		Broader GPIO, ADC, interfaces, and control options Smaller peripheral set; module-dependent pins

Use ESP8266 when a project mainly needs Wi-Fi. Use ESP32 when it also needs Bluetooth, richer control, or room to grow.

ONE FAMILY NAME · SEVERAL BOARD TYPES

The ESP32 family shares a name but not one exact feature set.

01

ESP32

Classic all-rounder with dual-core processing, Wi-Fi, and Bluetooth.

02

ESP32-S2

Wi-Fi-focused, single-core, with USB features and no Bluetooth.

03

ESP32-S3

Dual-core Wi-Fi and Bluetooth LE with extra acceleration for smart edge devices.

04

ESP32-C3

Low-cost single-core RISC-V option with Wi-Fi and Bluetooth LE.

05

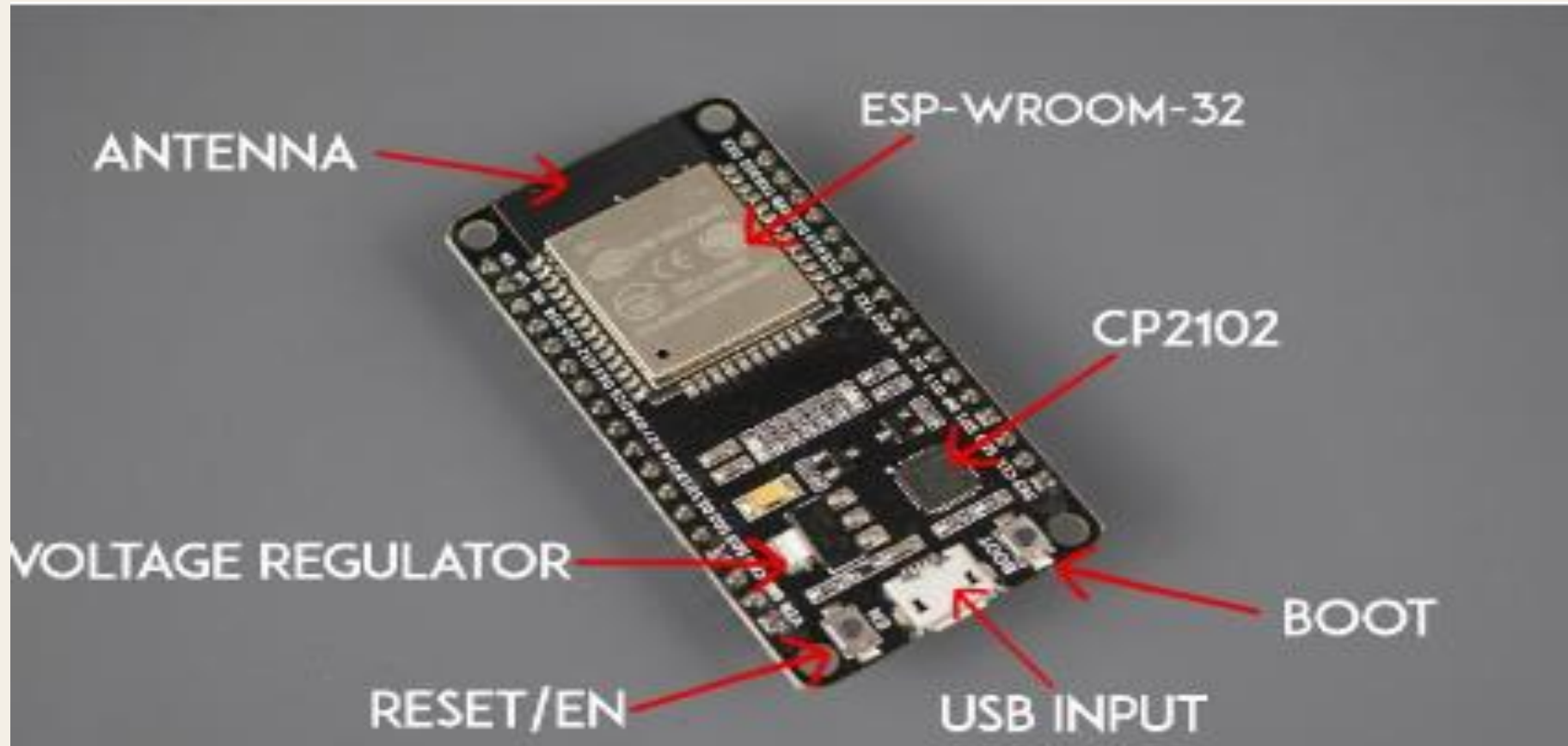
ESP32-C6

Newer low-power option with Wi-Fi 6, Bluetooth LE, and 802.15.4 support.

Before a first robotics project: check the exact board's pinout and voltage.

MEET THE BOARD

Read the development board before you connect it.



Pin placement varies by board. Confirm the exact pinout before wiring.

WHY THIS BOARD FITS THE PROJECT

Why ESP32 for robotics?



Wi-Fi

Monitor a robot, send sensor data, or control it from a network.



Bluetooth

Pair with a phone, gamepad, or nearby accessory.



Higher processing power

Coordinate sensing, decisions, and movement with more headroom.



More peripherals

Connect motors, sensors, displays, encoders, and communication modules.

One small board can combine local control with wireless communication.

A PRACTICAL DECISION RULE

Choose the board by the job, not by the name.

1

Simple 5 V control

Need the gentlest first step? Arduino UNO may be enough.

2

Connected and capable

Need Wi-Fi, Bluetooth, more processing, or more peripherals? ESP32 is a strong next choice.

3

Wi-Fi with low complexity

Need Wi-Fi at the lowest complexity and cost? ESP8266 may fit.

4

Check before wiring

Confirm the exact pinout, 3.3 V logic, power input, and boot controls.

ESP32 is powerful because it combines control, connectivity, and room to grow.